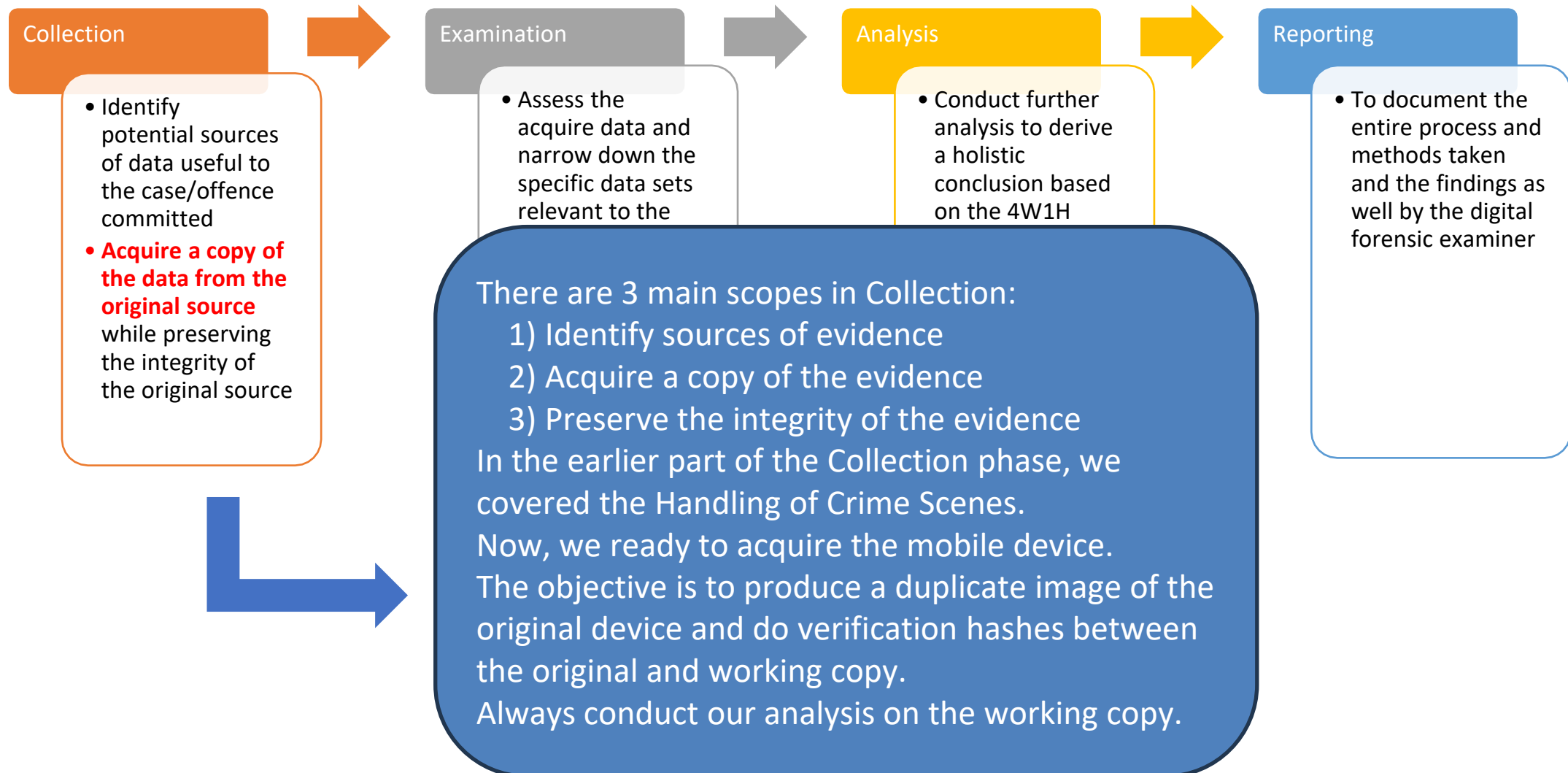


Week 3

Digital Forensic Framework: Collection (Acquisition)

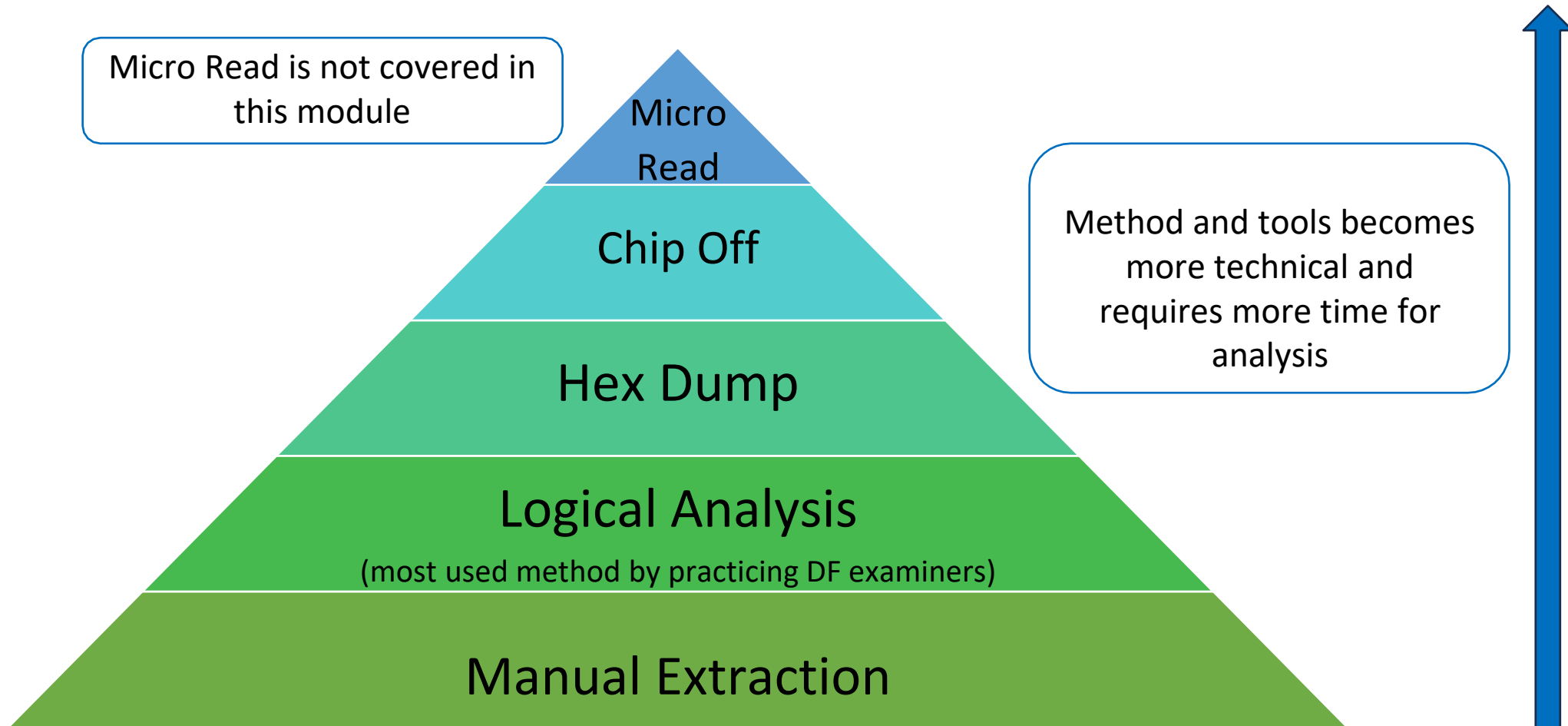
Digital Forensic Framework by NIST



Cellular Phone Tool Levelling Pyramid

- **Acquisition of mobile devices can be done via automated tools and manual effort.** Both automated and tools has their own sets of pros and cons. As a DF examiner, you can decide which tool suits best for your case scenario.
- The Cellular Phone Tool Levelling Pyramid was created to organise the different acquisition tools according to the technical effort and time required.

Cellular Phone Tool Levelling Pyramid



Manual Extraction

- This method simply involves **scrolling through the mobile device and view the data available on its screen**. Relevant data that can be potentially used as evidence are then photographed with a physical camera.
- The time needed to conduct manual extraction is very fast as it requires the least amount of technical effort by the DF examiner. Hence, this acquisition method can be done as soon as it is found in the crime scene at the beginning of the investigation.

Manual Extraction– Case Study

- You are involved in a police raid for a drug trafficking case. The location is the head boss's house. Upon forcible entry, the accused was seen to be in a call using his iPhone 12. You suspect that the iPhone 12 is the **main mobile device** used by the accused to communicate with his runners.

DF examiner performing a Manual Extraction:

After you have taken steps in data preservation, you navigated to his recent call logs in his iPhone 12 and photographically documented. In your expert opinion, these recent call logs can be considered as potential evidence as they could be contact numbers belonging to other runners involved in this drug trafficking operandi.

Logical Analysis

- Logical Analysis refers to **the extraction of the mobile device via an industry forensic tool**. There are two types of extractions: Logical and Physical.

Make & Model and Software Version

- Determines if logical and/or physical extraction can be performed on the mobile device
- Logical Extraction – only extracts **live data** from the mobile device
- Physical Extraction – extracts **both live and deleted data** from the mobile device

Data Storage Used

- Determines the time required to complete a mobile device extraction

Logical Analysis

- For the extraction of the mobile device via an industry forensic tool to be taken place successfully, there is there is a need to maintain an open communication between the device and forensic tool. Hence, **write blocker is not used in mobile acquisitions.**
- Open communication between the mobile device and the forensic tool can be enabled with the following state:
 - 1. Ensure the device is in powered on state when connected to the forensic tool.
 - 2. In Android devices, enable “USB Debugging” option
 - 3. In iOS devices, enable “Trust This Computer” option

Logical Analysis – Case Study

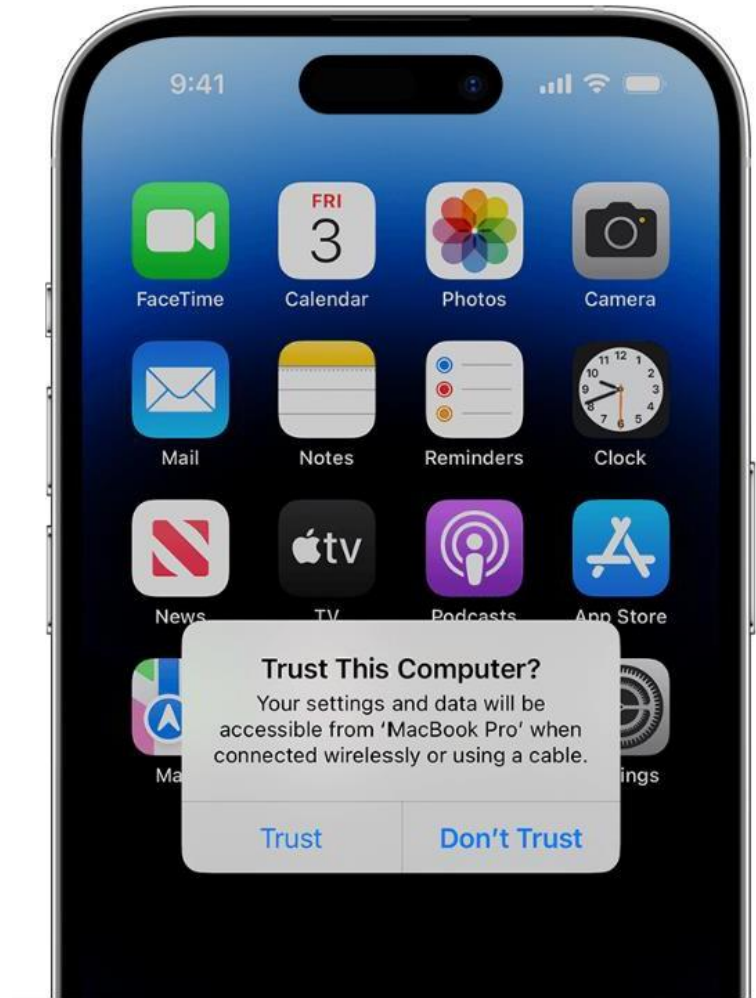
- You are involved in a police raid for a drug trafficking case. The location is the head boss's house. Upon forcible entry, the accused was seen to be in a call using his iPhone 12. You suspect that the iPhone 12 is the **main mobile device** used by the accused to communicate with his runners.

DF examiner performing a Logical Analysis:

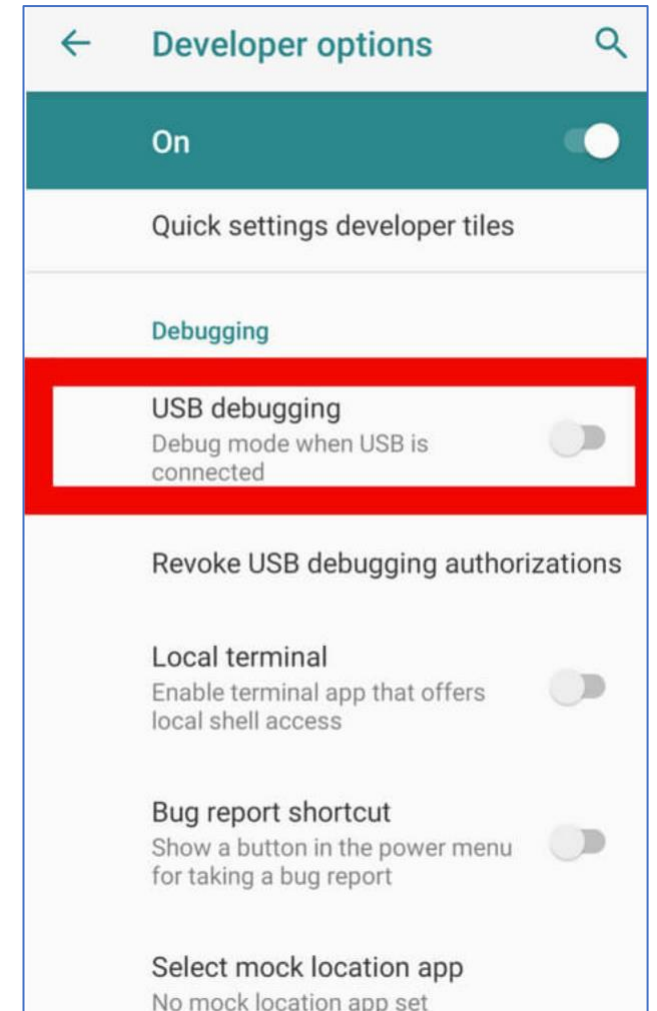
After you have taken steps in data preservation, you brought back the iPhone 12 device to the lab for an extraction. You plugged the device to Cellebrite Forensic Tool with write-block function disabled. Based on the make & model and the software version of the iPhone 12 device, the Cellebrite Forensic Tool can perform both logical and physical extractions.

You decided to do physical extraction to recover any potential deleted messages (SMSes, WhatsApp, Telegram), photos and videos that may be relevant to the case.

Logical Analysis



Enable this option in an iOS device



Enable this option in an Android device

Hex Dump

- A hex dump refers to the process of dumping the device's memory into a forensic workstation via a bootloader or unsigned code.
- The results of a successful hex dump is like a physical extraction done by a forensic tool. However, the output is in binary format. Hence, it takes more time and technical effort to analyse the output.
- **A DF examiner should only consider hex dump when physical extraction option is unavailable.**

Hex Dump – Case Study

- You are involved in a police raid for a drug trafficking case. The location is the head boss's house. Upon forcible entry, the accused was seen to be in a call using his iPhone 12. You suspect that the iPhone 12 is the **main mobile device** used by the accused to communicate with his runners.

DF examiner performing a Hex Dump:

After you have taken steps in data preservation, you brought back the iPhone 12 device to the lab for an extraction. You plugged the device to Cellebrite Forensic Tool with write-block function disabled. As the software version is not updated to the current version of that time, Cellebrite Forensic Tool is only able to perform logical extraction on the iPhone 12 device.

However, live data is not enough to support the case. And with **unavailability of physical extraction** the **hex dump** option, you have decided to perform a hex dump so you can access the device's deleted data.

Chip Off

- Chip off refers to **the acquisition of the actual memory chip** residing on the mobile device's motherboard. This method requires the act of desoldering of the memory chip from the motherboard and perform extraction of its data via a chip reader.

Chip Off – Case Study

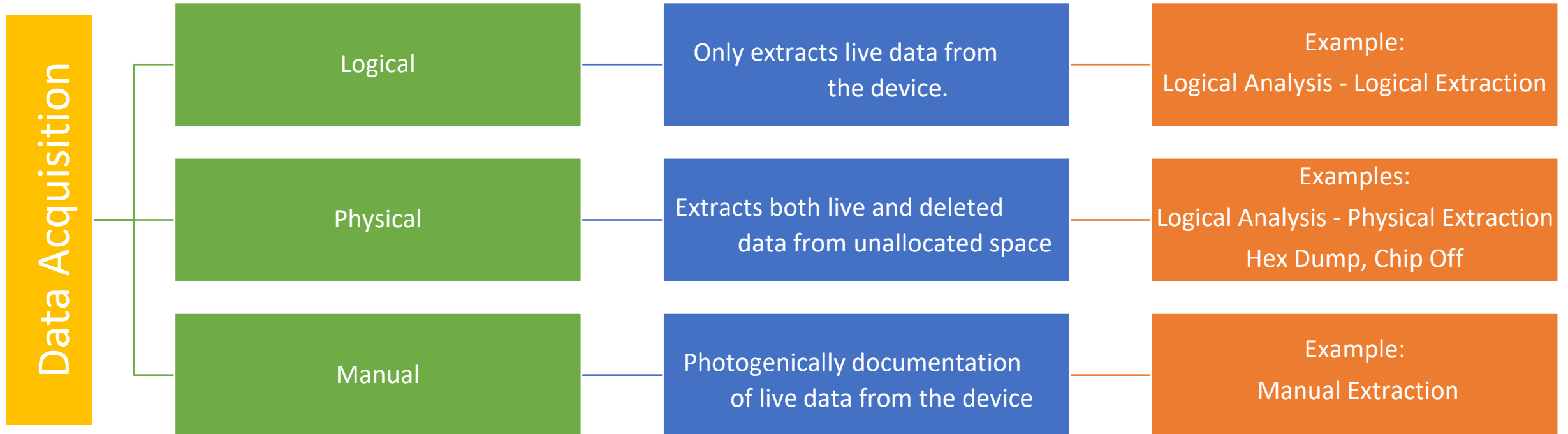
- You are involved in a police raid for a drug trafficking case. The location is the head boss's house. Upon forcible entry, the accused was seen to be in a call using his iPhone 12. You suspect that the iPhone 12 is the **main mobile device** used by the accused to communicate with his runners.

DF examiner performing a Chip Off:

At the crime scene, upon forcible entry, the accused panicked and quickly ran to the nearest window and threw his iPhone 12. You were able to retrieve the device, however, its damaged. Logical analysis and hex dump is not an option.

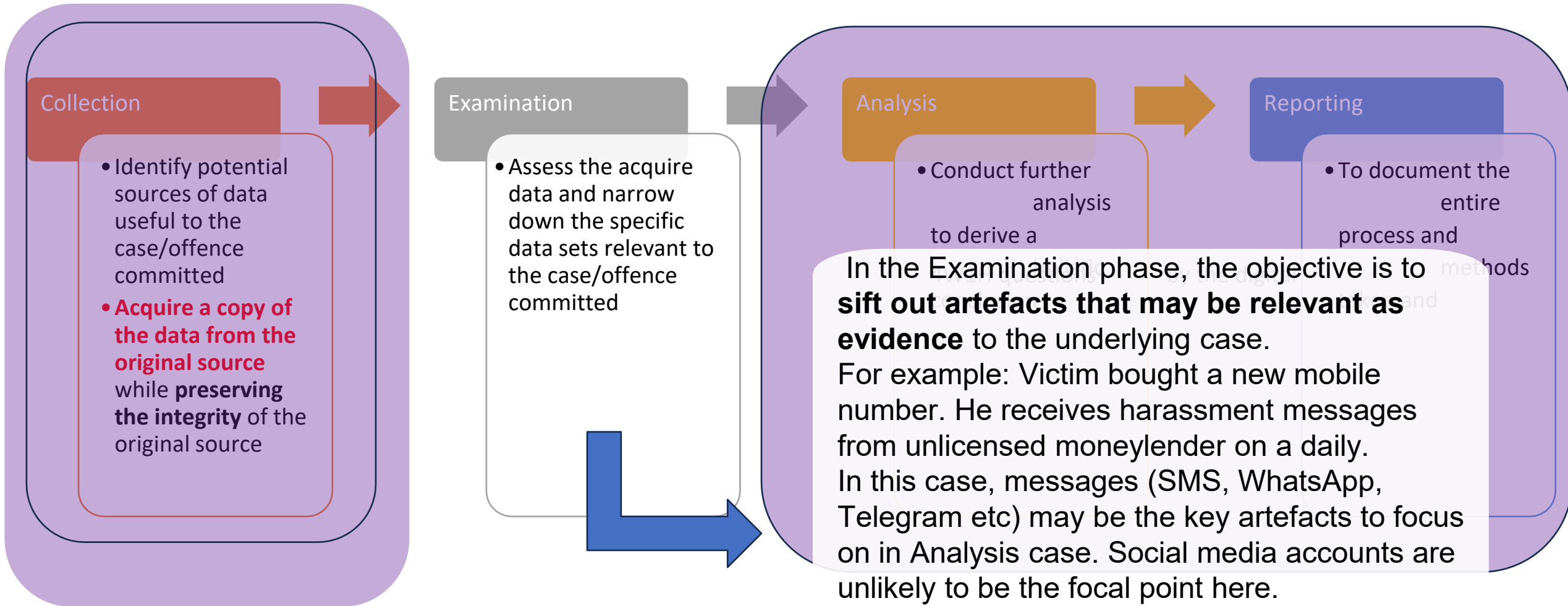
To extract any data, you need to access the device's memory chip directly. Hence, you perform the chip off method.

Data Acquisition Methods



Digital Forensic Framework: Examination

Digital Forensic Framework by NIST



Artefacts

SMSes

Third Party
Messaging
Applications

Call Logs

Installed
Applications

Media

Contact Book

Recently
Accessed
Locations

Recently
deleted media

Social Media
Accounts

Web History

Downloaded
Files

Notes

Other Potential Artefacts found in Mobile Devices

SD cards

- Artefacts: Media, Documents, Installed Applications
- Acquisition: Place the SD card in a card reader, attach to a write-blocker and acquire it via a computer forensic tool

SIM cards

- Artefacts: Contacts, Call Logs, SMS, MMS
- Acquisition: Place the SIM card in a SIM card reader and acquire it via a mobile forensic tool

Next Learning Topics in Week 5

- Types of Analysis
 - Temporal
 - Relational
 - Functional
- Report Writing
 - Key information to be included in the report